

Lab6: Logistic Regression and Metrics

DataLab

2026.10.01

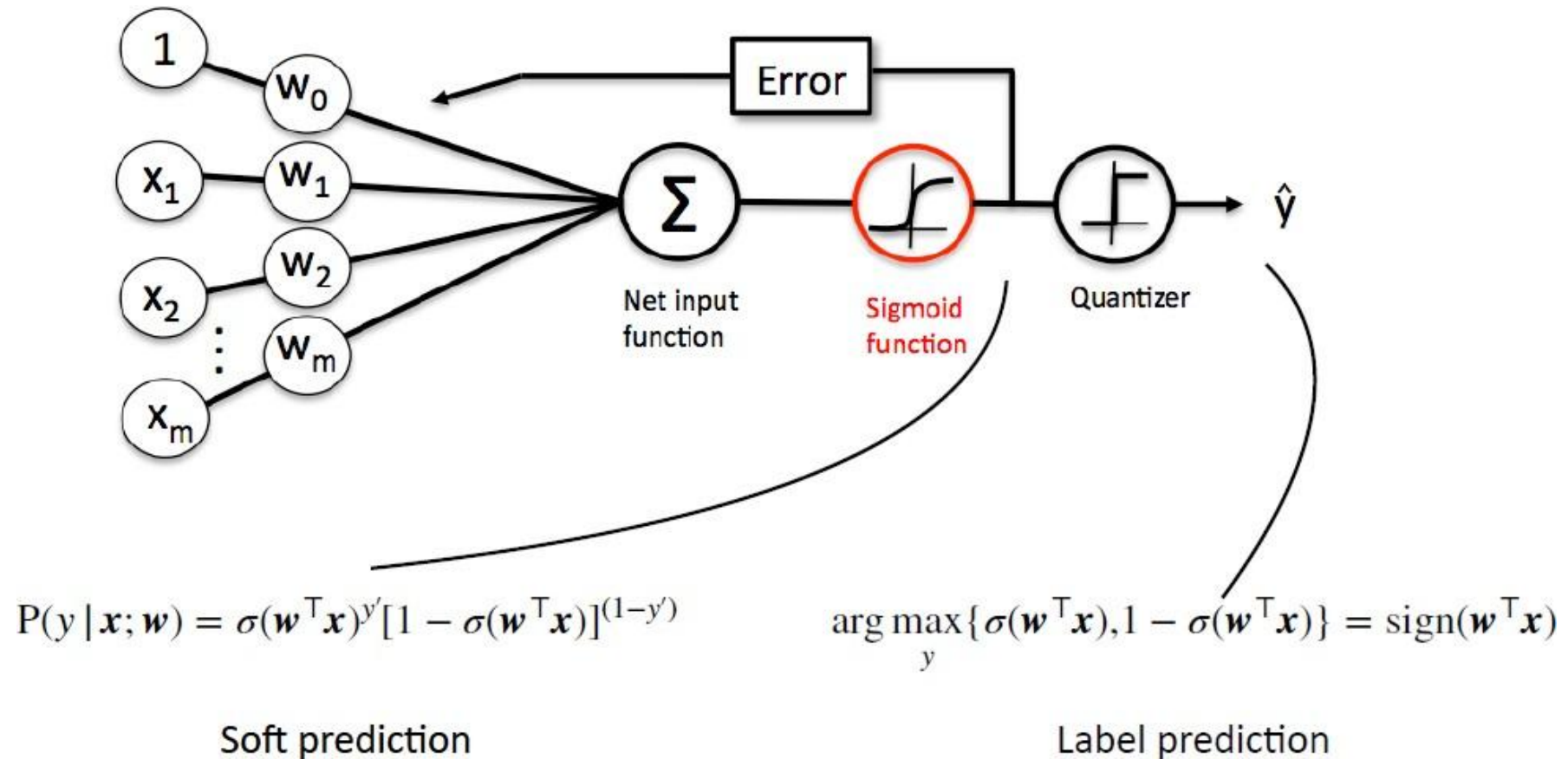
Outline

- Brief Review: Logistic Regression
- Common Evaluation Metrics for Binary Classification
 - Confusion Matrix
 - Soft Classifiers – ROC Curve

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Logistic Regression



Logistic Regression + Regularization

- 把 regularization term 加到 loss 內，讓模型在學 weights 的時候，會傾向選擇比較簡單的模型。

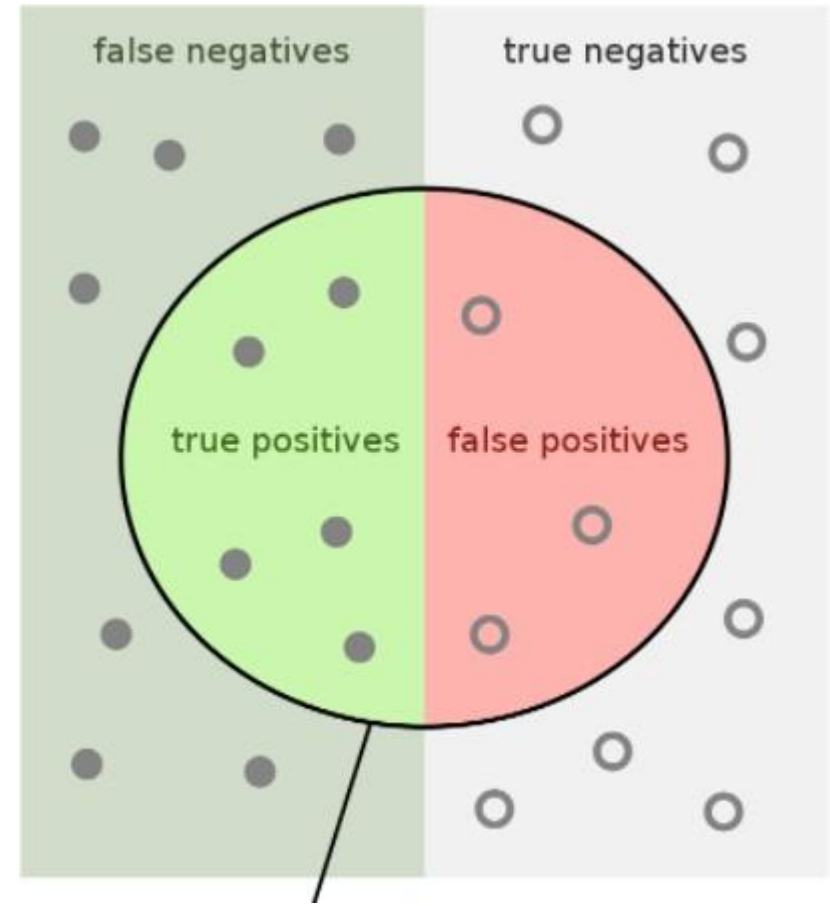
$$\arg \max_w \log P(\mathbb{X} | w) - \frac{\alpha}{2} \|w\|^2$$

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Confusion Matrix

- It is important to know how the model make wrong prediction
- In **binary classification**, confusion matrix is a common tool to analyze the predictions



Confusion Matrix

- It is important to know how the model make wrong prediction.
- e.g. 檢測絕症
 - 寧可讓多一點人到 TP & FP，也不要讓 FN 很高
 - i.e. TPR higher

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

		Predicted class	
		P'	N'
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

Confusion Matrix

- Precision(PRE) & Recall Rate(REC)

$$PRE = \frac{TP}{TP + FP}, \quad (\text{the higher, the better})$$

$$REC = \frac{TP}{TP + FN} = TPR. \quad (\text{the higher, the better})$$

- F-1 Score

$$F_1 = 2 \frac{(PRE * REC)}{PRE + REC}, \quad (\text{the higher, the better})$$

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ROC Curve

- ROC curve analyze the performance for **every threshold in soft classifiers**

- In X-axis $FPR = \frac{FP}{FP + TN}$

- In Y-axis $TPR = \frac{TP}{TP + FN}$

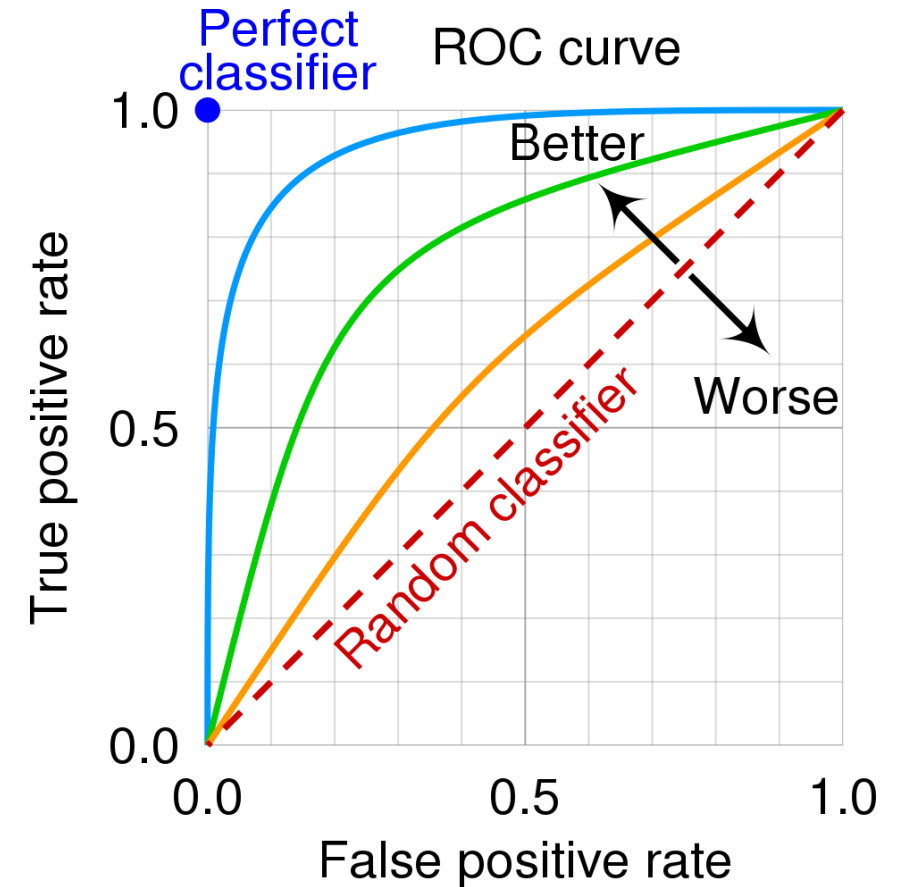
1	
1	
0.87	θ
0.64	$\Downarrow$
$\vdots$	
-0.88	
-0.93	
-1	

ROC Curve

- ROC curve analyze the performance for **every threshold in soft classifiers**

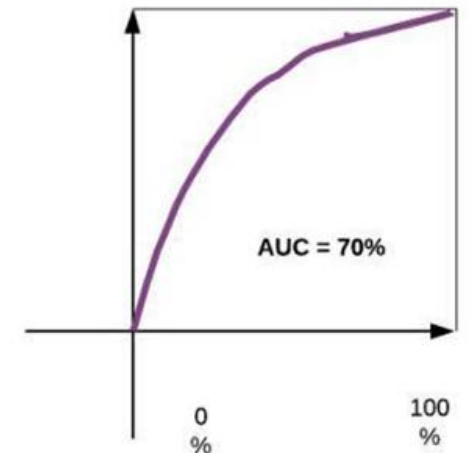
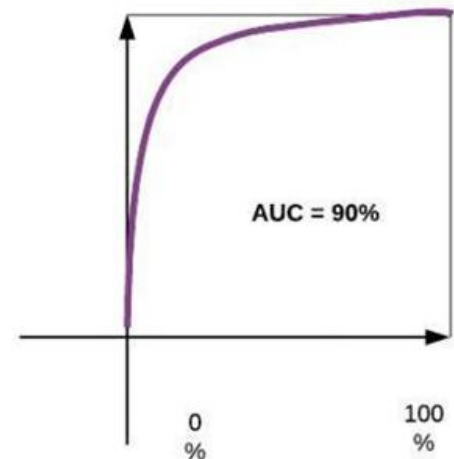
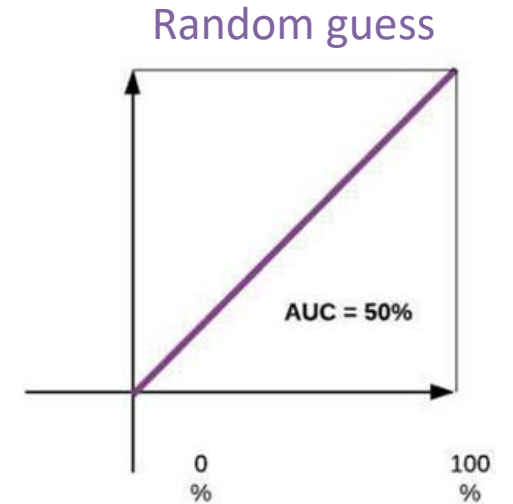
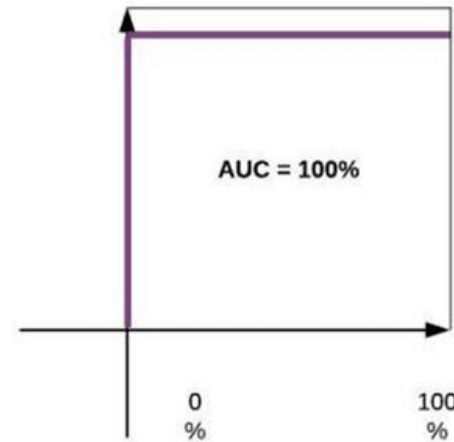
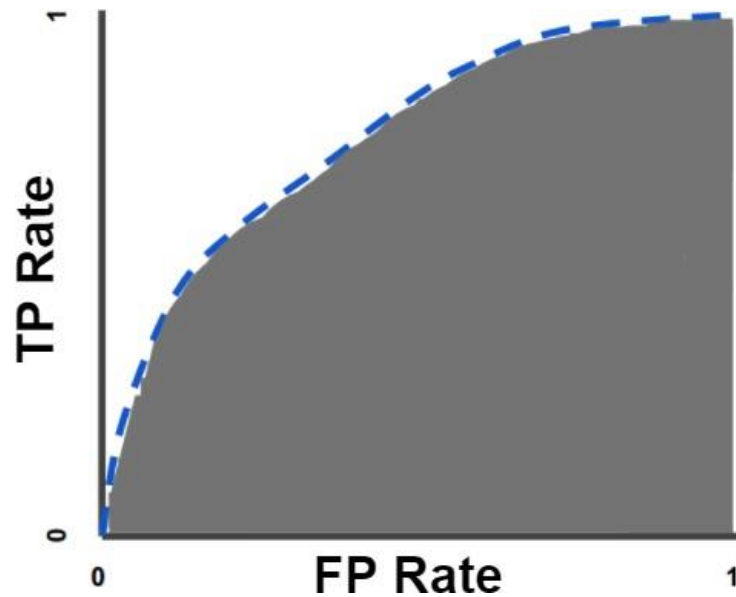
- In X-axis $FPR = \frac{FP}{FP + TN}$

- In Y-axis $TPR = \frac{TP}{TP + FN}$



AUC

- AUC (Area Under the ROC Curve)
 - ROC can be quantified using AUC



Homework

- Homework: Lab06
 - Lab06: Logistic Regression, Metrics

Reference

- https://bookdown.org/ccwang/medical_statistics6/section-43.html
- https://bookdown.org/ccwang/medical_statistics6/bernoulli.html
- https://bookdown.org/ccwang/medical_statistics6/binomial.html
- https://bookdown.org/ccwang/medical_statistics6/likelihood-definition.html
- https://en.wikipedia.org/wiki/Sensitivity_and_specificity
- <https://commons.wikimedia.org/w/index.php?curid=109730045>
- <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>
- <https://medium.com/acing-ai/what-is-auc-446a71810df9>
- https://github.com/dariyasydykova/open_projects/tree/master/ROC_animation